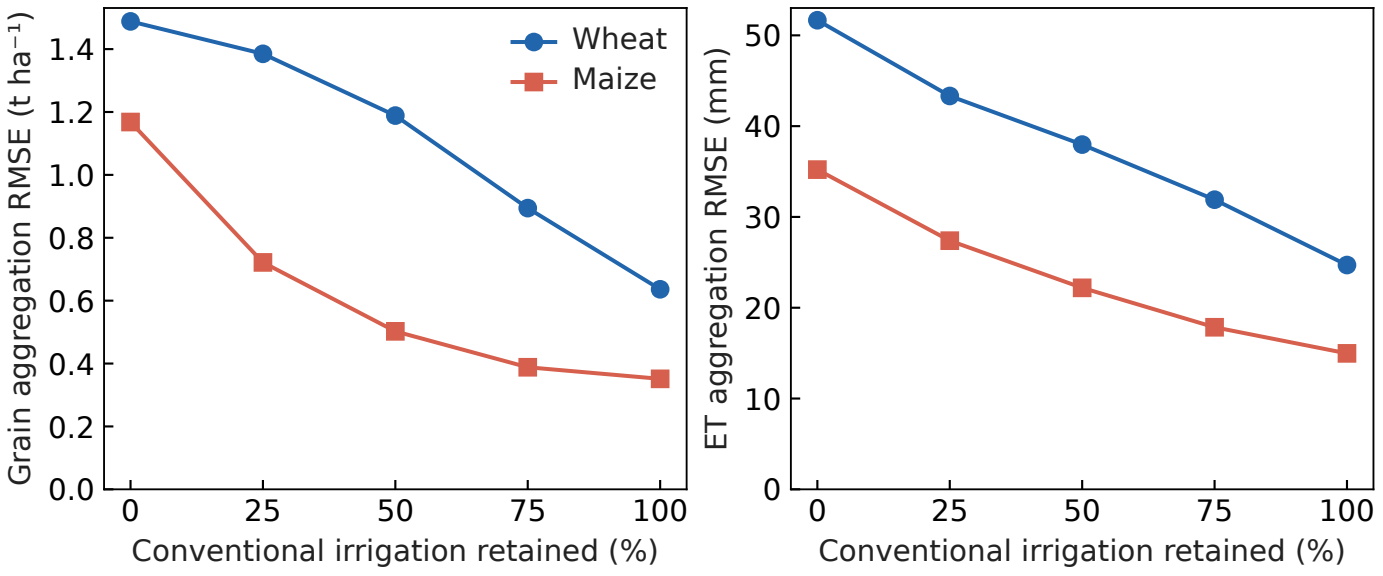
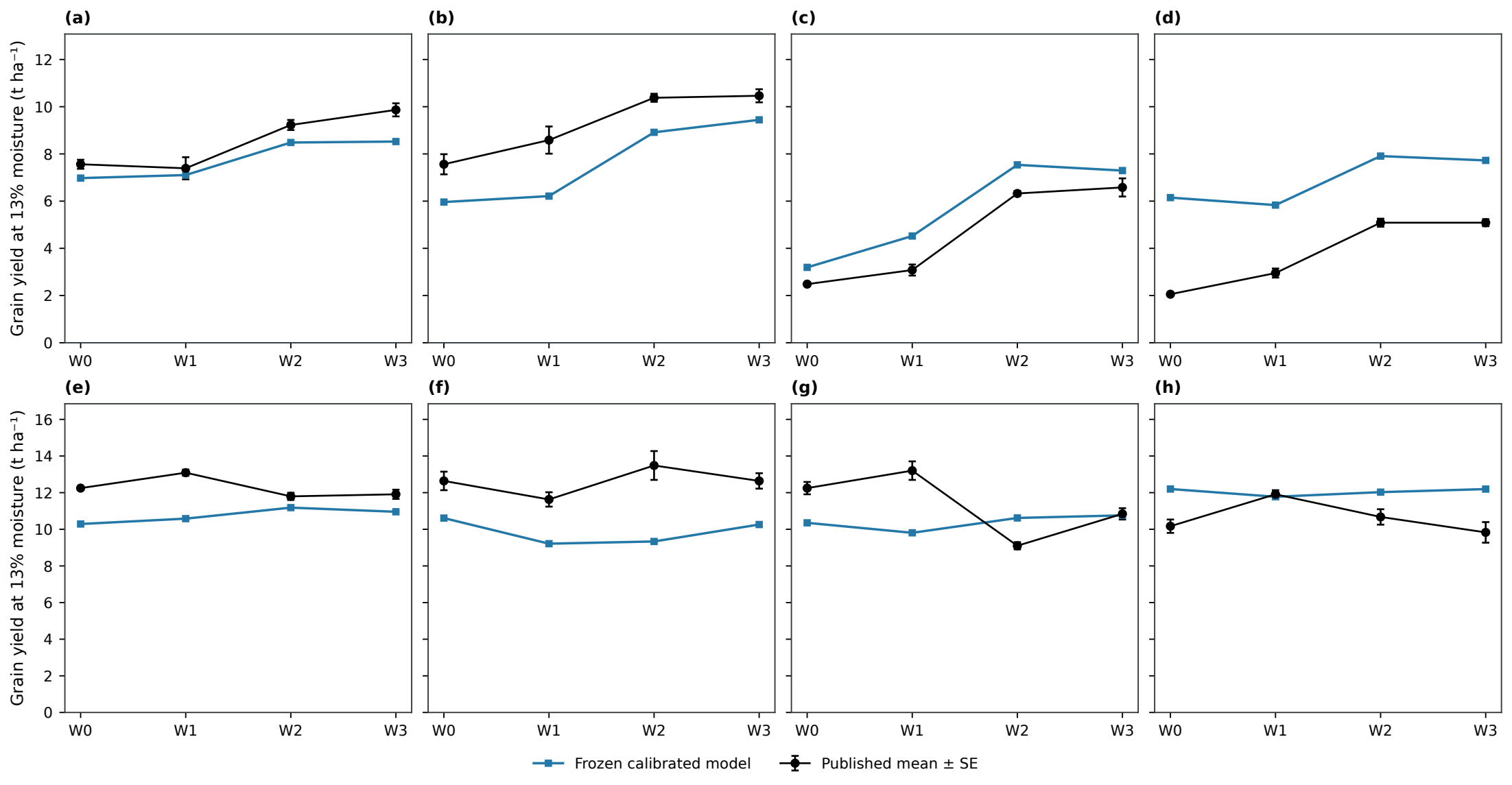
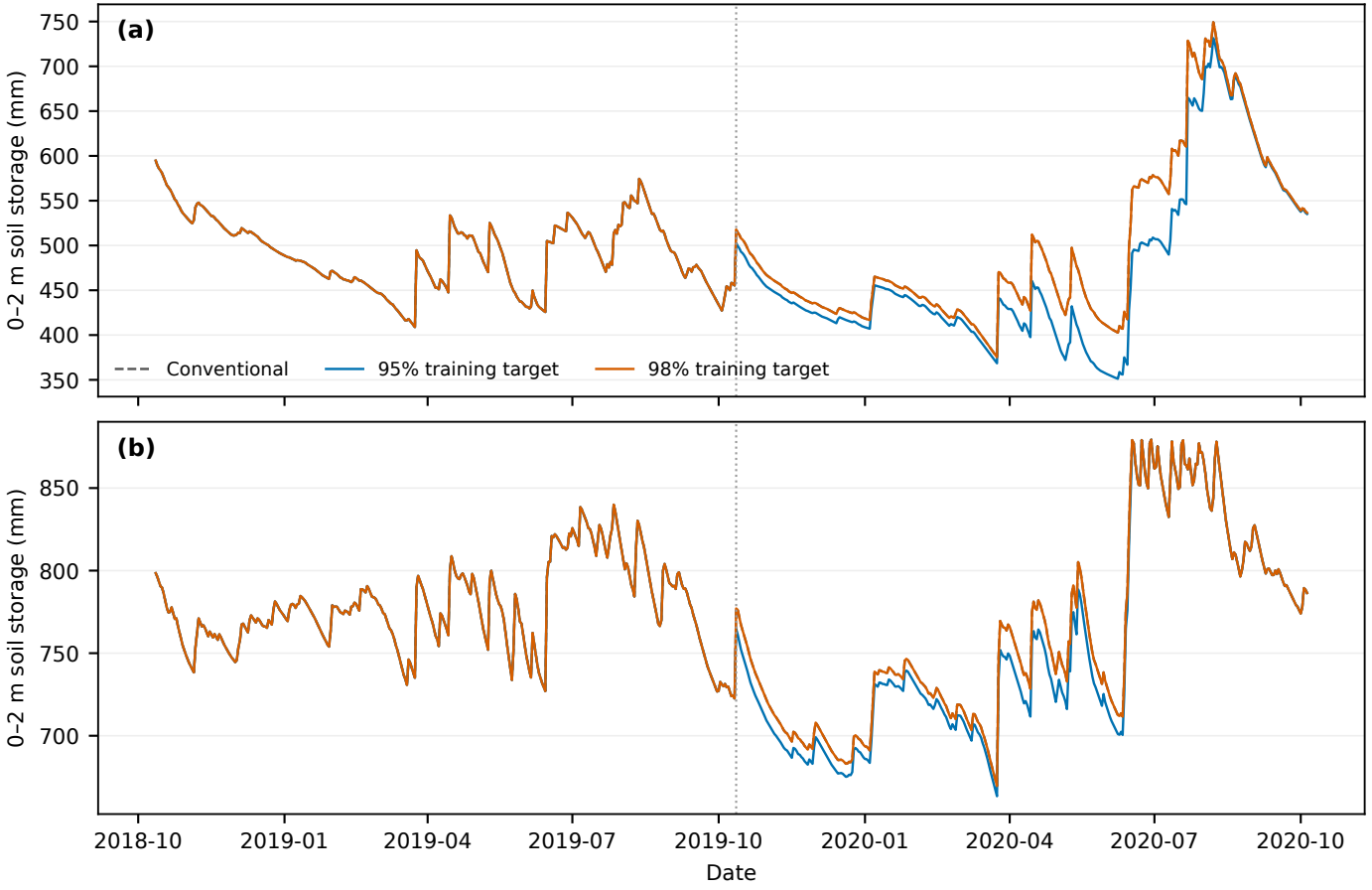
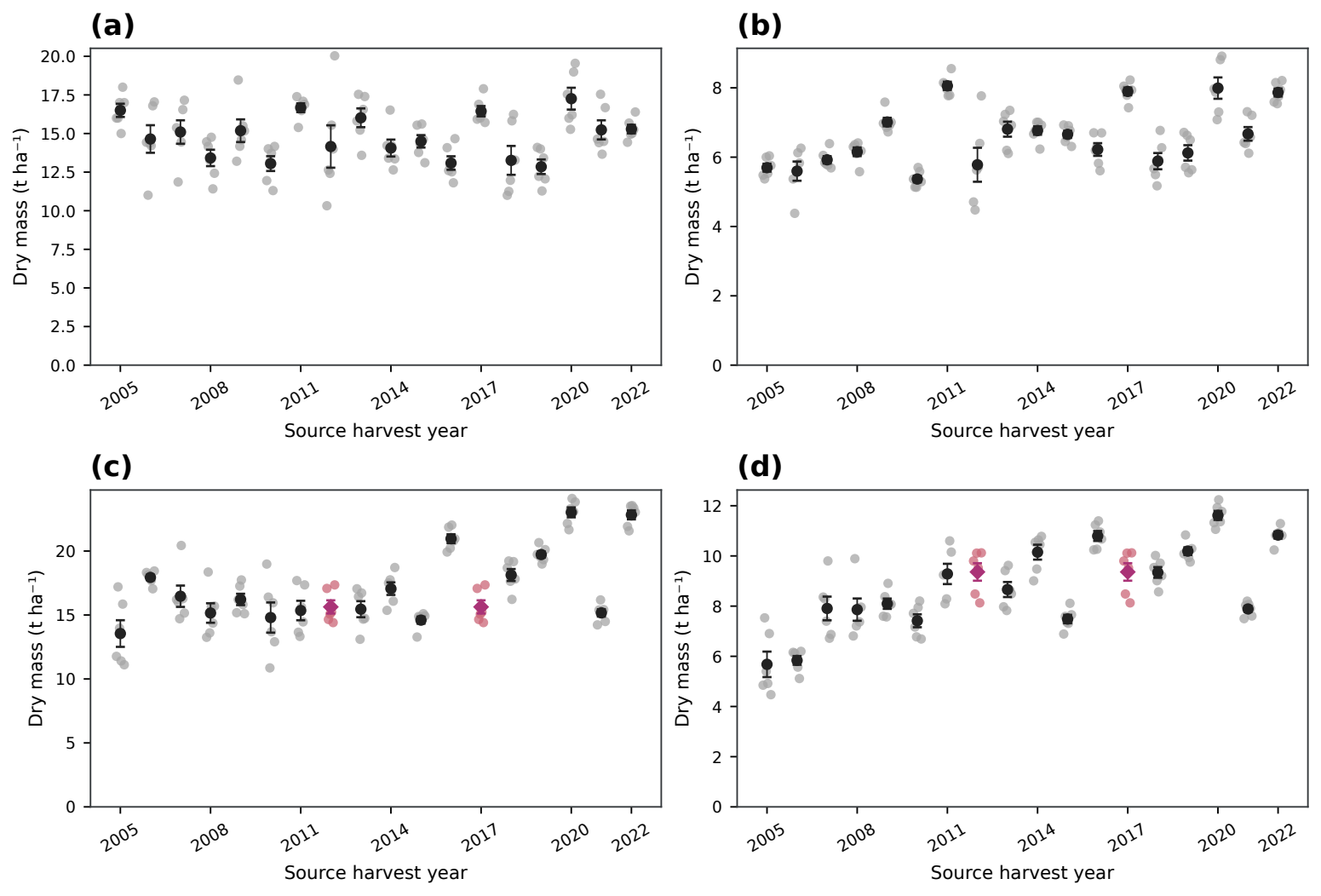
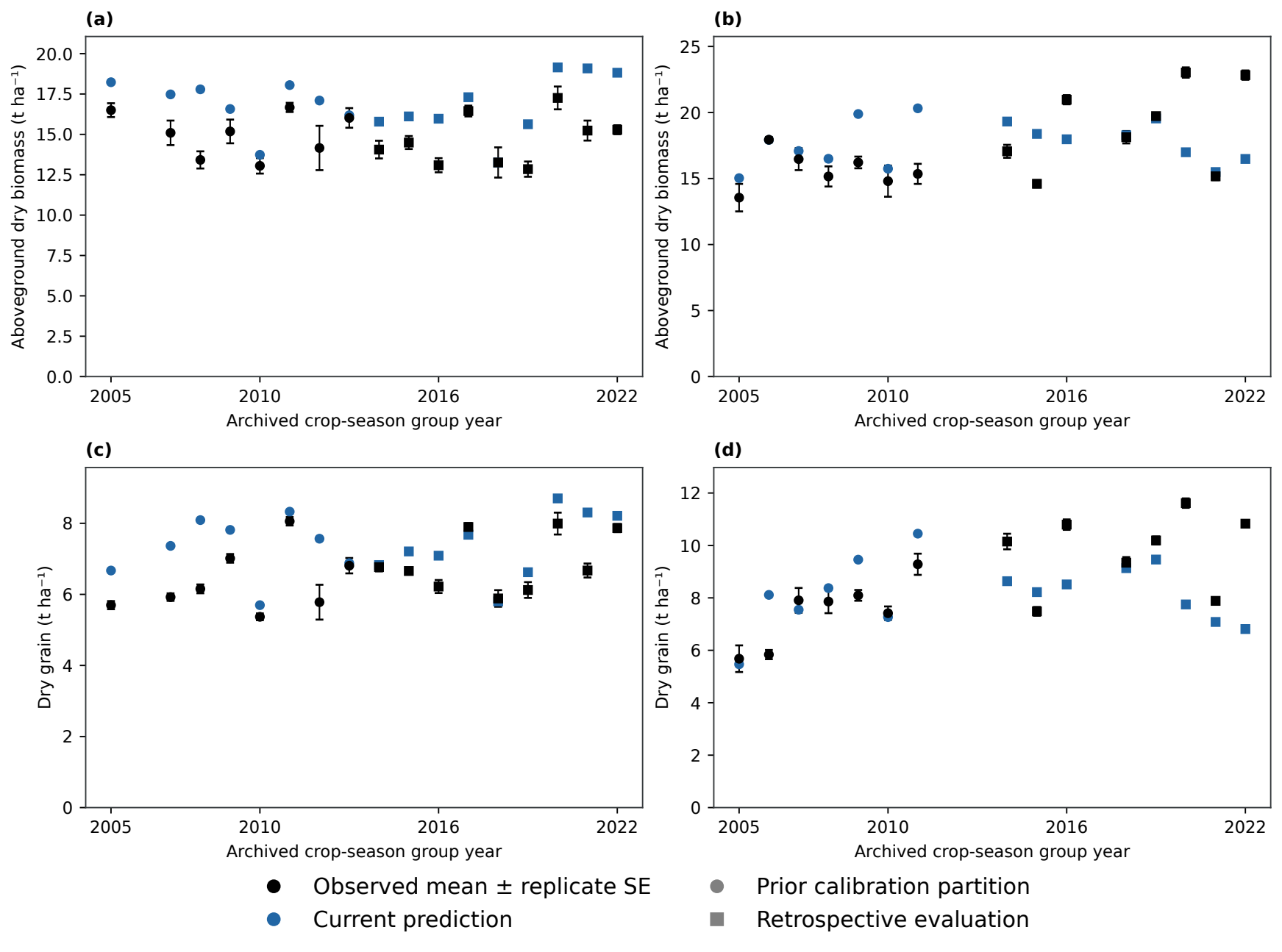
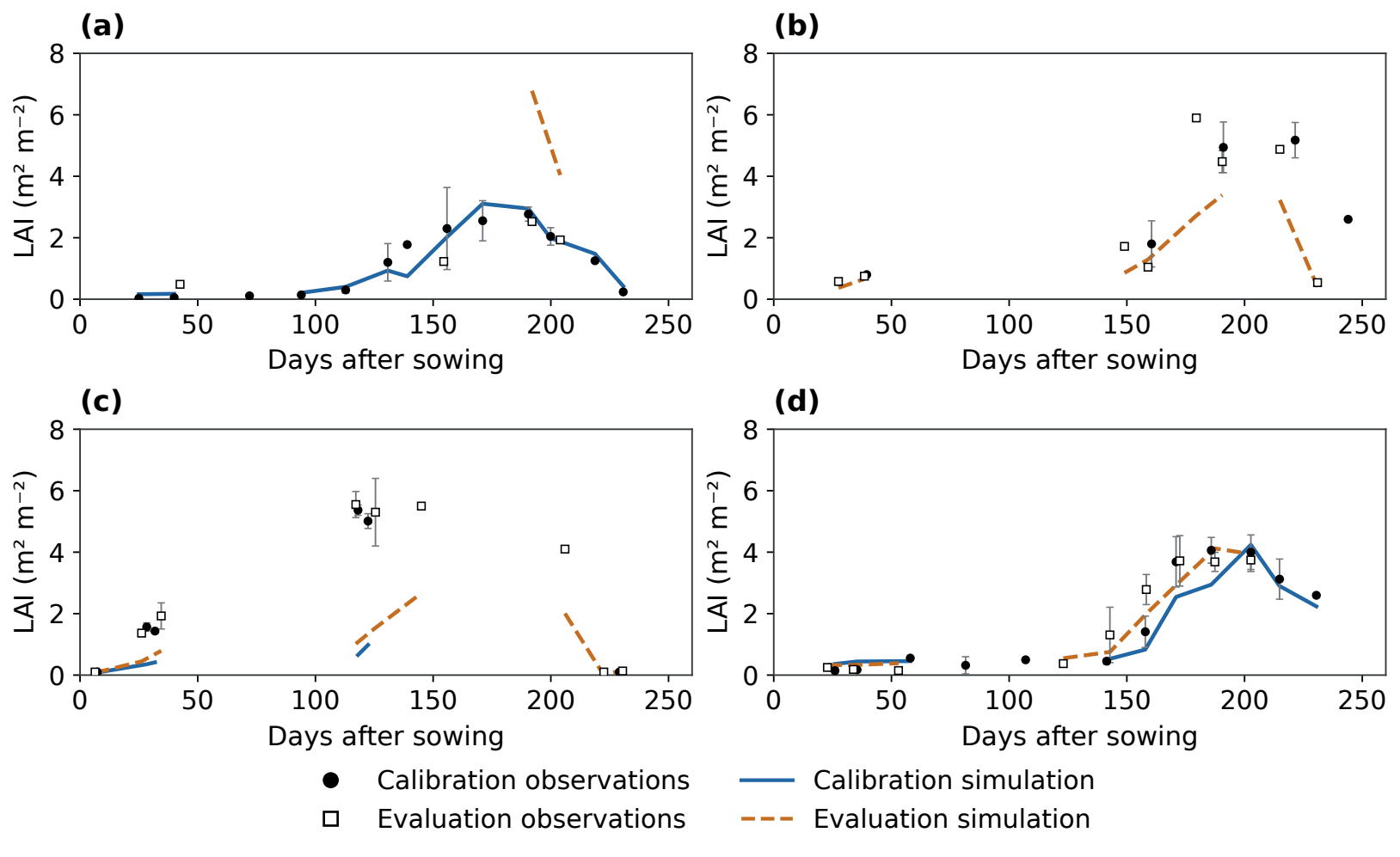

Wheat
Maize
Grain aggregation RMSE (t ha⁻¹)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
25
50
75
100
Conventional irrigation retained (%)
ET aggregation RMSE (mm)
0
10
20
30
40
50
0
25
50
75
100
Conventional irrigation retained (%)

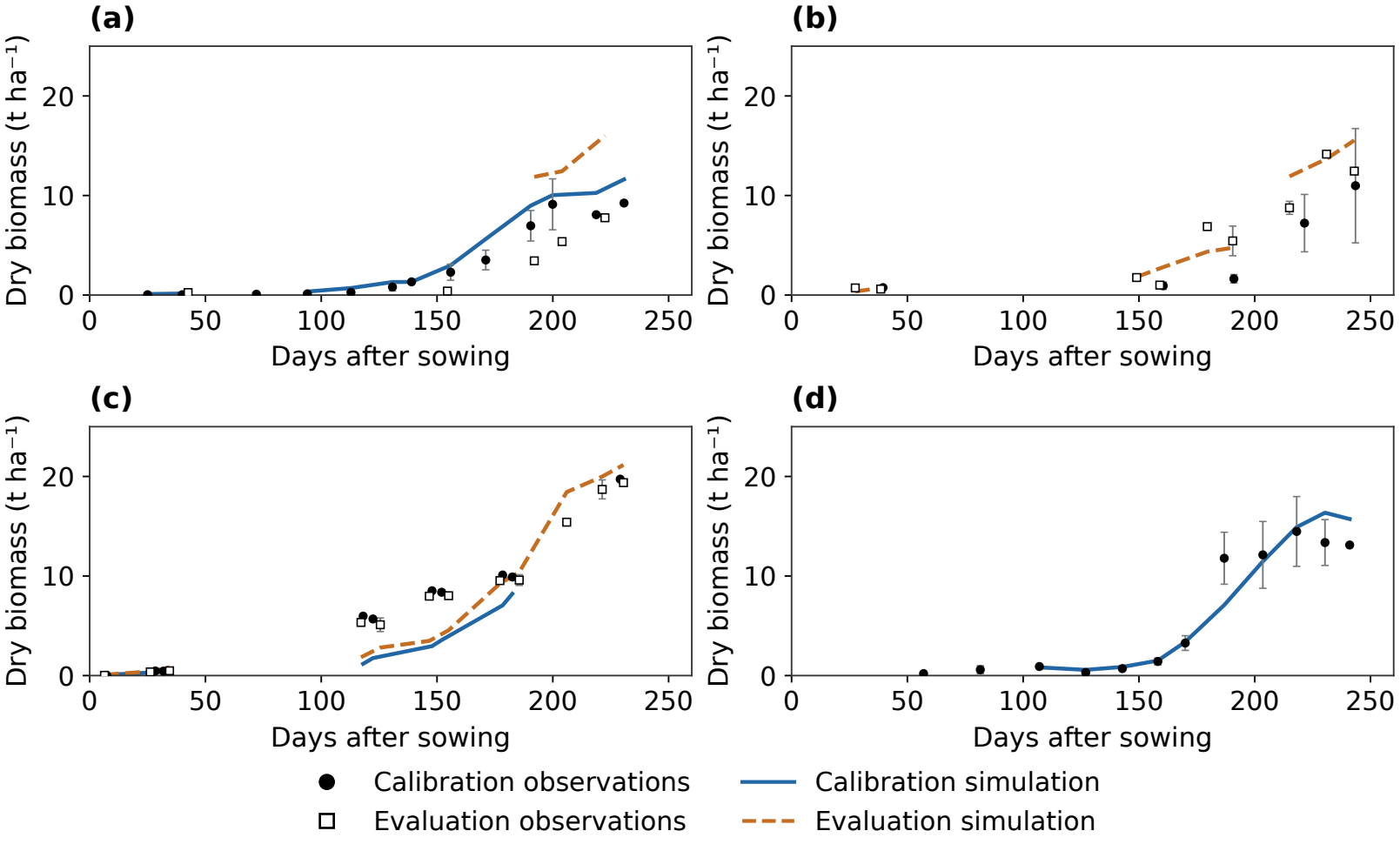

(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
Grain yield at 13% moisture (t ha⁻¹)
W0
W1
W2
W3
Frozen calibrated model
Published mean ± SE

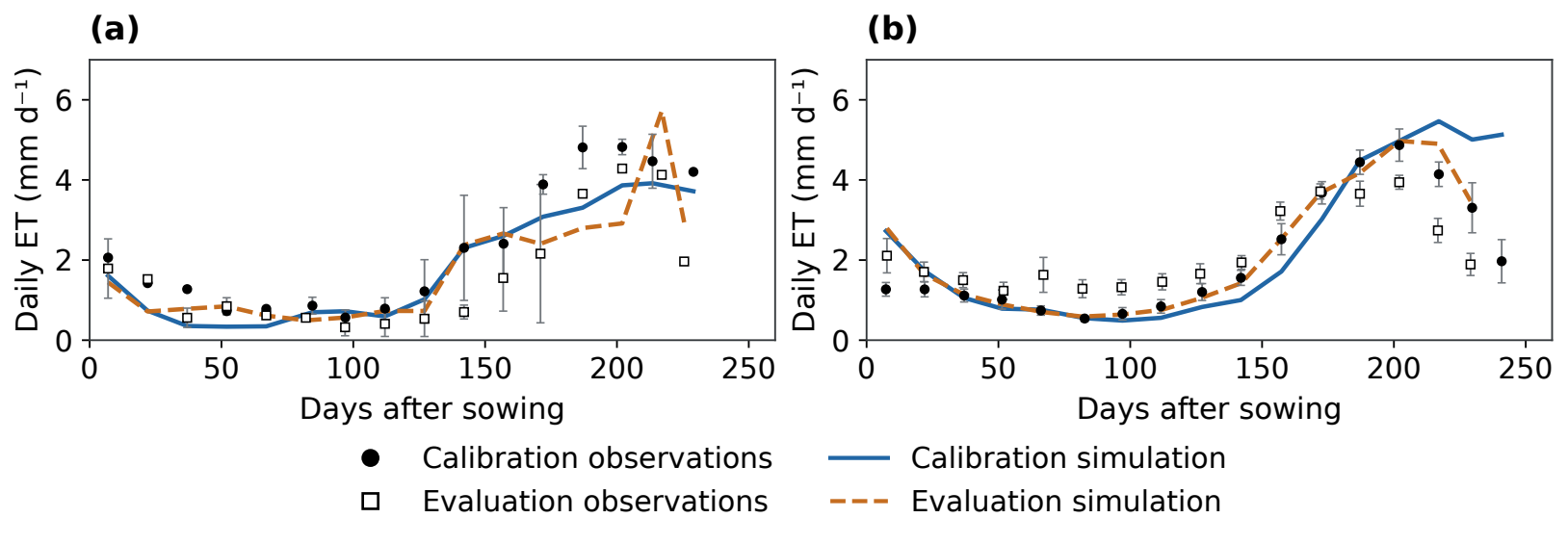

(a)
0–2 m soil storage (mm)
750
700
650
600
550
500
450
400
350
Conventional
95% training target
98% training target
(b)
0–2 m soil storage (mm)
850
800
750
700
2018-10
2019-01
2019-04
2019-07
2019-10
2020-01
2020-04
2020-07
2020-10
Date

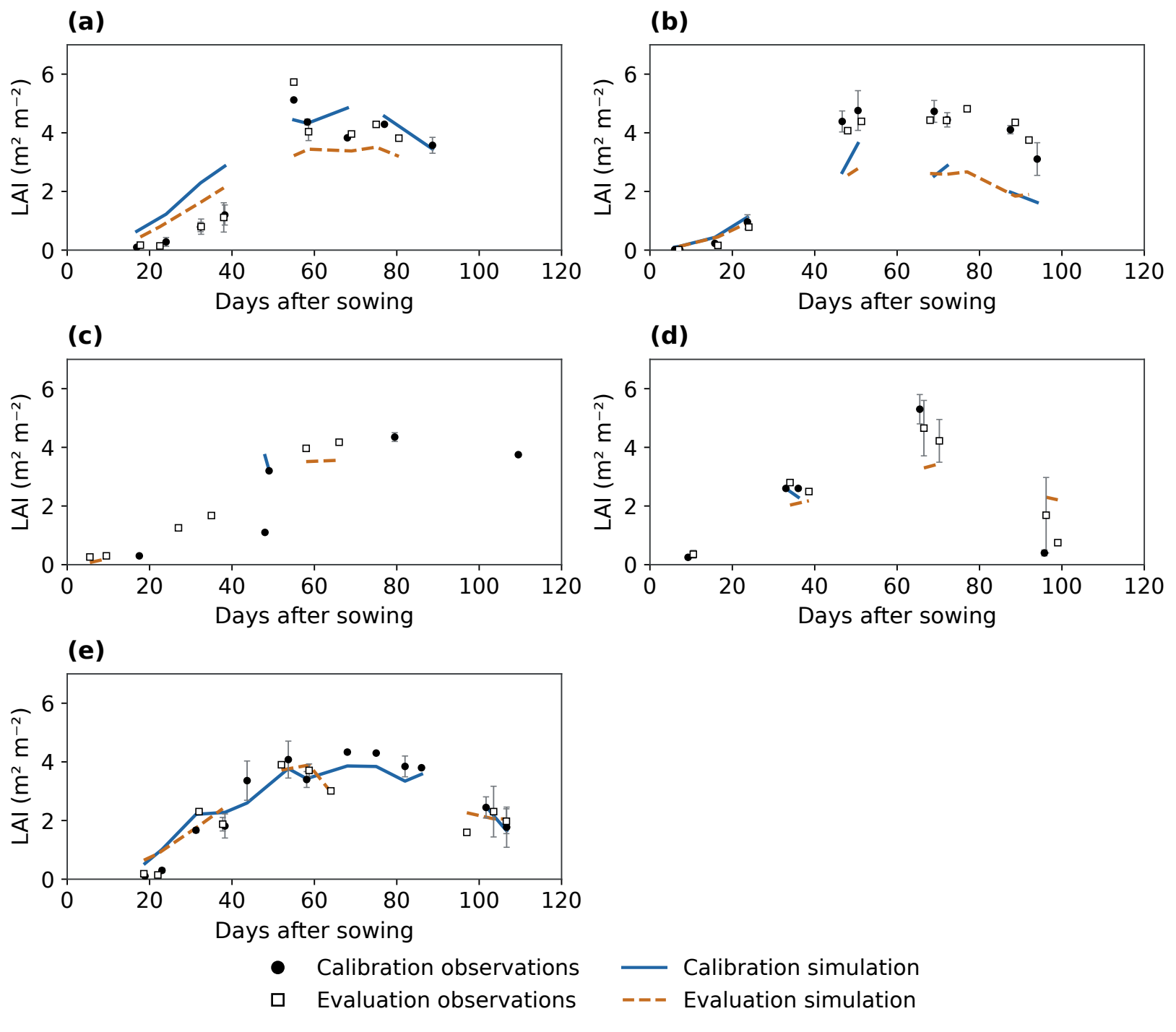

(a)
(b)
(c)
(d)
Dry mass (t ha⁻¹)
Source harvest year
2005
2008
2011
2014
2017
2020
2022

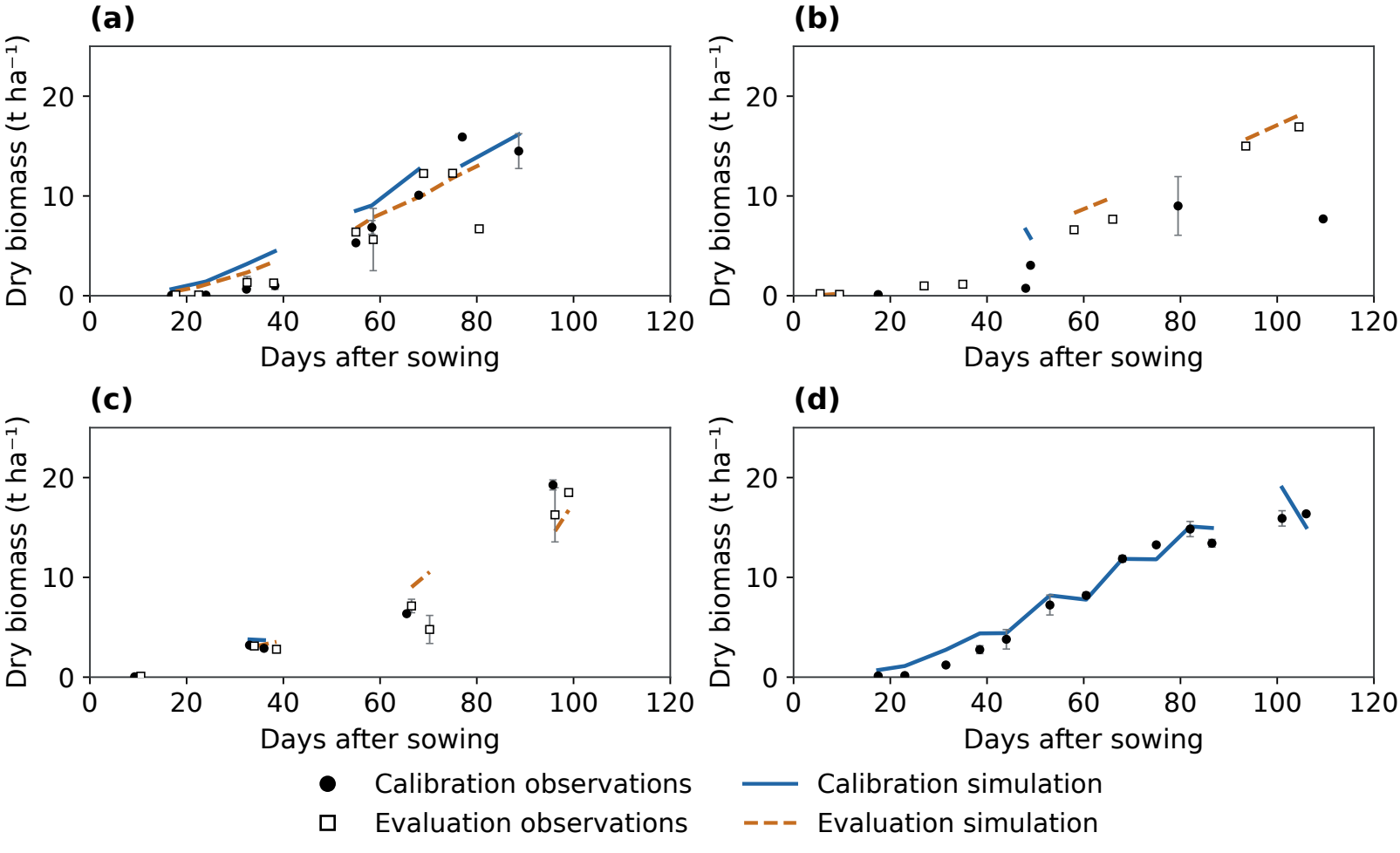

(a)
(b)
(c)
(d)
Aboveground dry biomass (t ha⁻¹)
Dry grain (t ha⁻¹)
Archived crop-season group year
2005
2010
2016
2022
Observed mean ± replicate SE
Current prediction
Prior calibration partition
Retrospective evaluation

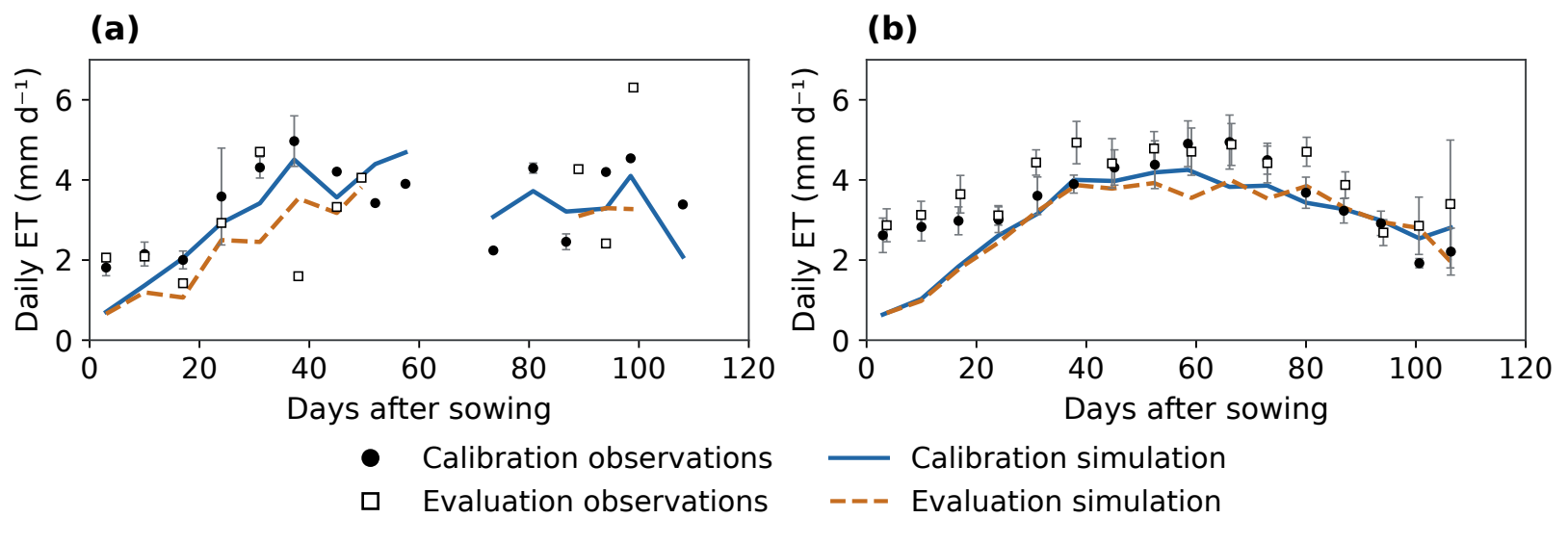

(a)
(b)
(c)
(d)
LAI (m² m⁻²)
Days after sowing
0
2
4
6
8
0
50
100
150
200
250
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
(b)
(c)
(d)
Dry biomass (t ha⁻¹)
Days after sowing
0
10
20
0
50
100
150
200
250
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
(b)
Daily ET (mm d⁻¹)
Days after sowing
0
2
4
6
0
50
100
150
200
250
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
(b)
(c)
(d)
(e)
LAI (m² m⁻²)
Days after sowing
0
2
4
6
0
20
40
60
80
100
120
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
(b)
(c)
(d)
Dry biomass (t ha⁻¹)
Days after sowing
0
10
20
0
20
40
60
80
100
120
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
(b)
Daily ET (mm d⁻¹)
Days after sowing
0
2
4
6
0
20
40
60
80
100
120
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation